Rishi Roy

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Professional Summary

Computer Science Engineering student focused on **backend development, AI/ML, and full-stack applications**. Experienced in building REST APIs, AI-powered systems, authentication workflows, and machine learning applications using **Python, FastAPI, React, SQLAlchemy, and AWS**. Hands-on experience developing end-to-end applications through industry internship, hackathons, research, and software projects. Strong foundation in **software engineering, databases, machine learning, and API development**.

Skills

AI & ML	Machine Learning, Data Preprocessing, Model Training & Evaluation, Computer Vision, Emotion Classification, scikit-learn
Backend	Python, FastAPI, Flask, Node.js, REST APIs, SQLAlchemy, JWT Authentication
Frontend	React.js, HTML, CSS, Streamlit, Tailwind CSS
Databases	PostgreSQL, SQLite, SQL
Languages	Python, Java, C++, JavaScript
Tools & Cloud	AWS, Git, GitHub, Postman

Experience

API Research & Integration Intern — DNAG Technologies 3 Months

- Researched **Flipkart Seller APIs** covering seller onboarding, product listing, shipment management, and order processing.
- Tested and validated REST APIs using **Postman**, verifying authentication, requests, responses, parameters, and endpoint behavior.
- Prepared technical documentation for API endpoints and their integration into business workflows.

Education

The Oxford College of Engineering 2023 – 2027
B.E. in Computer Science and Engineering **(CGPA: 9.12/10)** *Expected 2027*

Achievements

- **1st Place** — Oxy-Hack 18-Hour Hackathon, The Oxford College of Engineering.
- **2nd Place** — HackSetu (Theme-1) at TATVA'26, BNM Institute of Technology.
- **Research Publication** — Published *"A Simplified Approach to Direct AC-AC Matrix Converter Using Verilog HDL for Real-Time Application"* in **IEEE Xplore (2025)**.

Projects

TalentAI HRMS — AI-Powered Human Resource Management System GitHub

- Built a full-stack **AI-powered HRMS** covering **5 core HR functions**: recruitment, employee management, attendance, payroll, and performance tracking.
- Automated **2 recruitment workflows** — AI resume screening and interview question generation — targeting repetitive screening tasks; research involving **100 HR professionals and recruiters in India** found AI-based resume screening significantly associated with **time-to-hire**.
- Implemented backend services using **FastAPI, SQLAlchemy, SQLite, and JWT authentication** for secure and structured data management.
- Developed responsive **React.js** dashboards with role-based access and analytics to centralize HR operations and reduce manual workflow overhead.

SAHAJ — Smart Adaptive Highway & Junction GitHub

- Developed a full-stack **crowd and traffic management system** that predicts real-time crowd severity levels to intelligently coordinate traffic signals.
- Designed a **3-tier architecture** combining **Spring Boot, FastAPI, and React.js** for backend services, ML processing, and visualization.
- Integrated a dedicated **machine learning service** through FastAPI to generate crowd severity predictions from traffic data.
- Built a **PostgreSQL-backed dashboard** to monitor crowd density and traffic insights across multiple junctions.